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James B. Leonard, Kaitlyn Brown & Jonathan M. Colvin

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COMMENTARY



Enhancing U.S. poison center surveillance with verbatim substance upload

James B. Leonard^a , Kaitlyn Brown^b  and Jonathan M. Colvin^c 

^aWashington Poison Center, Seattle, WA, USA; ^bAmerica's Poison Centers, Arlington, VA, USA; ^cDrug & Poison Information Center, Cincinnati Children's hospital Medical Center, Cincinnati, OH, USA

ABSTRACT

Introduction: America's Poison Centers National Poison Data system® is a widely respected databased crucial for identification of evolving drug and toxic exposure trends. This database has been used for real time surveillance for about 25 years. Surveillance has historically been heavily dependent on coding, but updates including the addition of verbatim substance to the database can enhance the surveillance tool.

Methods: We queried the National Poison Data System® for exposures including kratom from January 1, 2023 through July 31, 2025 and reviewed the uploaded verbatim substance to identify exposures of 7-hydroxymitragynine. We compared this to the exposures identified by product code.

Results: There were 186 exposures to 7-hydroxymitragynine. Most ($n=149$) were identified by a product code, but an additional 37 had originally been associated only with kratom and were identified as 7-hydroxymitragynine by substance verbatim. Additionally, 7-hydroxymitragynine was identified by verbatim in November of 2024 whereas the product code was implemented in February 2025.

Discussion: Reviewing uploaded substance verbatim was able to identify 7-hydroxymitragynine exposures earlier (November 2024 versus February 2025) and captured an additional 20% of exposures.

Conclusion: The addition of substance verbatim field, when combined with other tools may help enhance surveillance performed using the National Poison Data System®.

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7-hydroxymitragynine; surveillance; US Poison Center; verbatim; epidemiology

Introduction

America's Poison Centers National Poison Data System® (NPDS) is a widely respected poison exposure database that is crucial for identification of evolving drug and other toxin exposure trends [1,2]. The NPDS comprises exposures reported to the 53 regional poison centers that cover the United States, Virgin Islands, Puerto Rico, Guam, Federated States of Micronesia, and American Samoa. Exposures are managed clinically by Specialists in Poison Information and coded data are uploaded in near real-time. Data collection in NPDS began in 1983 and transitioned to a web-based platform in 2000 to conduct near real-time surveillance for incidents of public health concern [3]. Real-time surveillance is performed by looking for anomalies in geographic case volumes or case-based definitions. Historically, the data available in NPDS has been limited to the coded data uploaded from Poison Centers. Analysis and signal detection using NPDS relies on

nearly 500,000 Micromedex Product Codes (product codes) to detect an outbreak with a specific product, and over 1,100 America's Poison Centers Generic Codes® to detect an outbreak in a broader substance category [4]. Despite the extensive number of product and generic codes, new substances without assigned codes may delay the identification of new substance trends of concern.

In response to the rapidly changing substance exposure landscape, the NPDS was expanded to enable the free-text substance verbatim to be uploaded from all U.S. Poison Centers in November of 2024 and required starting January 2025, but only partially compliant. The substance verbatim is intended to capture the name of the substance or product exactly as it was reported by a caller (i.e., "a small brown hairy spider"). The substance verbatim field can be used to identify new products and trends in new product exposures before a product or generic code is implemented.

Recent attention has been given to 7-hydroxymitragynine, an opioid agonist that is an alkaloid of kratom [5]. In response to an increase in kratom exposures and news around 7-hydroxymitragynine, a product code was added to the database for 7-hydroxymitragynine in February 2025 [5]. From February 1, 2025 through April 30, 2025, there were 53 exposures reported to the NPDS with that product code [5].

This small number of exposures likely underrepresents the case volume, especially immediately after introduction of a new product code. There may not be consistent notification to frontline staff which may result in incomplete case capture, thus evaluation of substance verbatims may help identify additional exposures. We describe a use-case scenario using verbatims to identify additional 7-hydroxymitragynine exposures that did not have the correct product code in the early phases of this outbreak and identify exposures occurring before the addition of the product code.

Methods

We performed a retrospective review of NPDS from January 1, 2023 through July 31, 2025 with the America's Poison Centers Generic Code® for kratom and reviewed the substance verbatim field to identify additional exposures to 7-hydroxymitragynine not captured by a product code or assigned to an incorrect code. These exposures were queried on August 18, 2025.

We included single- or multi-substance, closed human exposures that included the kratom generic code as at least one substance. Exposure dates, information about the substance(s) involved by product and generic codes, and the substance verbatim field were analyzed. The substance verbatim field was isolated and each product was searched online to determine the active ingredients. All three authors reviewed the results of the search and independently determined if the product was a 7-hydroxymitragynine product. If the substance verbatim field documented 7-hydroxymitragynine or a similar derivation (e.g., 7-hydroxy, 7-OH) that was not associated with a specific brand, reviewers classified it as 7-hydroxymitragynine. Disagreements were resolved through discussion. Names not specific enough to be determined were considered not 7-hydroxymitragynine. Exposures coded as 7-hydroxymitragynine, regardless of substance verbatim were considered to be 7-hydroxymitragynine as the substance.

At the time of the query, 32 centers were uploading verbatim terms. Throughout the covered period, as few as 1 center was uploading which increased over time; there were 2 centers in July 2024 (as part of a

pilot project), 22 centers in January 2025, and 30 centers in July 2025. We did not query by individual centers and center information was not provided with the dataset. We limited our analysis to descriptive statistics. This project was determined to be exempt by the institutional review board at Cincinnati Children's Hospital Medical Center.

Results

There were 4,436 exposure cases associated with the kratom generic code reported over the period from all US Poison Centers. Of these exposure cases, 960 included a substance verbatim value; 158 of these verbatim values were unique (e.g., "kratom" or "7-hydroxymitragynine"). Most substance verbatims (n=763) did not provide additional information about the product (e.g., the verbatim was "kratom") and 67 were not 7-hydroxymitragynine, with a total of 186 exposures to 7-hydroxymitragynine. Most were identified by a 7-hydroxymitragynine product code (n=149) and 37 were identified only by the substance verbatim field (Figure 1). The earliest exposure identified as a coded product was reported mid-February 2025, but the earliest exposure identified by a substance verbatim value was November 2024. Figure 2 shows the epidemic curve using product codes (gray) and substance verbatims (black) divided into 7-day intervals along with the number of centers uploading verbatim over time.

Discussion

We were able to identify the onset of an epidemic of 7-hydroxymitragynine earlier using verbatim terms compared with product codes alone. This was identified about 3 months before the addition of the 7-hydroxymitragynine code and we identified an additional 37 exposures (20%) using the verbatim compared to the product code alone. We queried the database in mid-August, after the U.S. Food and Drug Administration had put out press releases about 7-hydroxymitragynine on July 29, 2025. It is possible that some poison centers had gone back and re-coded exposures that had initially been coded as "kratom" to "7-hydroxymitragynine." If this is true, then using substance verbatim would have been even more valuable early in an outbreak.

The NPDS® is a powerful tool for tracking the epidemiology of poisoning, especially once an incident of public health significance has been identified. Prior to the upload of the substance verbatim field, identification and tracking of new substances was limited by the availability of an assigned product or generic code. For

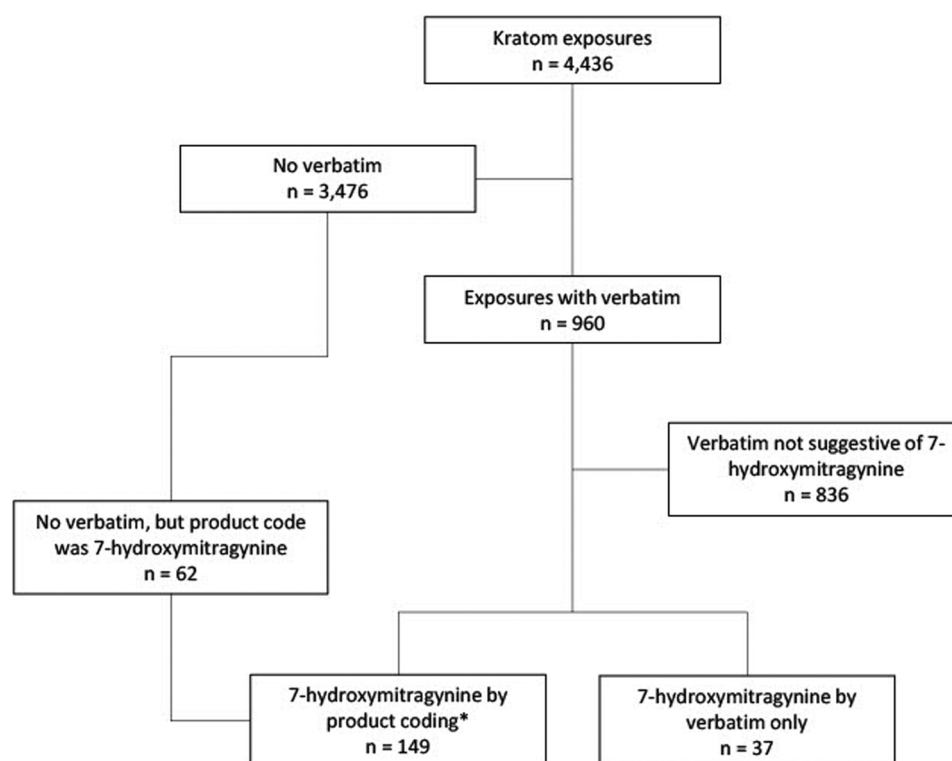


Figure 1. Flow diagram of kratom exposures. Flow diagram of kratom exposures describing how exposures were allocated and reviewed. *7-hydroxymitragynine by coding includes some exposures that also had verbatim suggestive of 7-hydroxymitragynine.

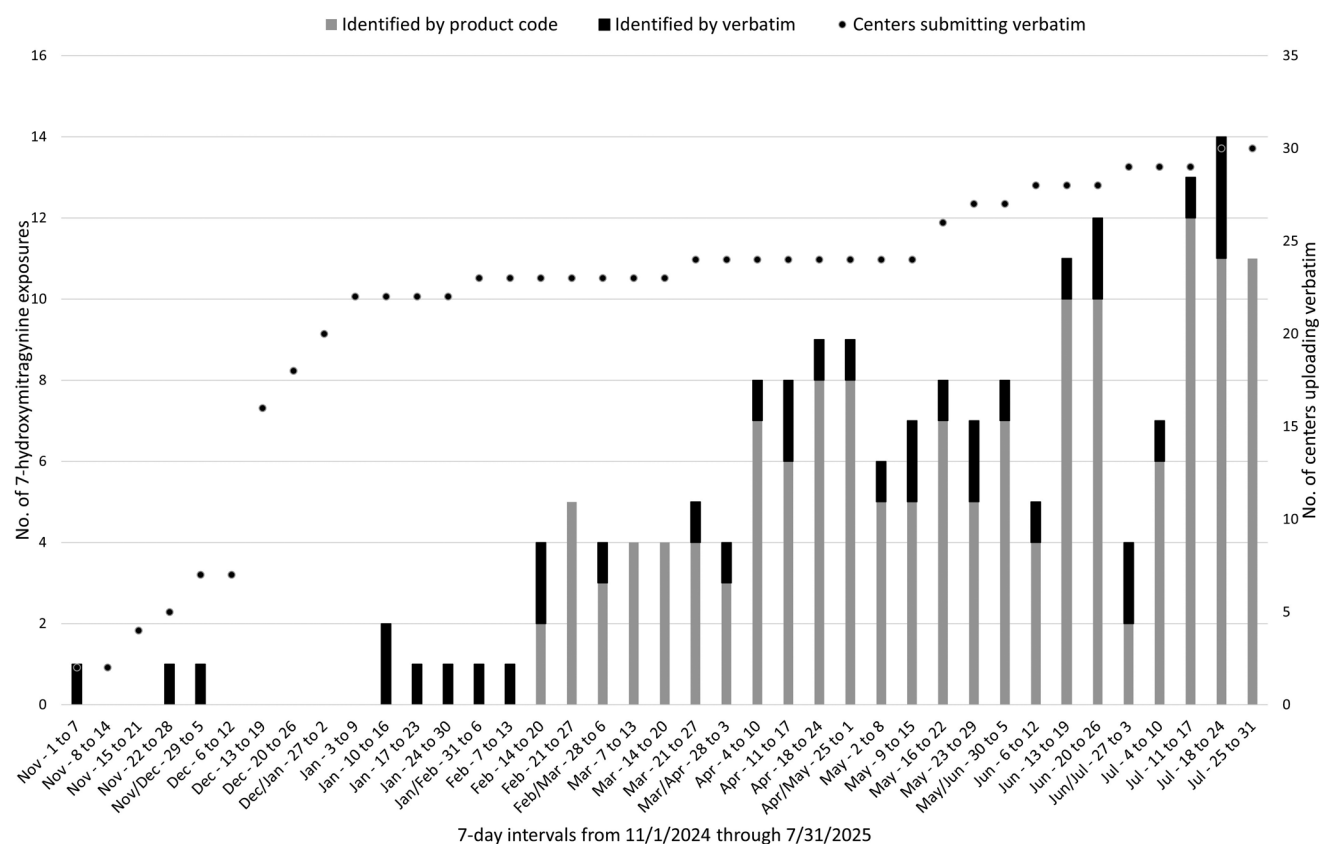


Figure 2. Epidemic curve of 7-hydroxymitragynine exposures reported to U.S. Poison Centers. Epidemic curve of 7-hydroxymitragynine exposures reported to US poison centers with exposures identified by product code in gray and those identified by verbatim substance in black columns. The number of centers contributing substance verbatims is demonstrated with a black dot on the right y-axis. Each column is a 7-day interval starting at November 1, 2024 and ending July 31, 2025.

example, the Diamond Shruumz™ outbreak in the summer of 2024 was identified by email communications between Poison Centers after only one or two severe cases. Cases had to be manually sent to America's Poison Centers and recoded with an emergent product code that was implemented within 24h. Case identification and tracking improve over time as America's Poison Centers® Rapid Coding Committee is able to activate emergent product codes and Poison Centers integrate the new product codes into their work flow.

Limitations

Poison center consultation is optional and exposures do not represent the total epidemiology of use. There is reporting bias, and factors like uncertainty with medical management of new products, media attention, and legislative actions may result in an increase in calls. Importantly, approximately half of the poison centers were uploading substance verbatim during the period of the outbreak, further limiting our search. As more centers upload verbatims, surveillance with this tool may improve. Additionally, some poison specialists may have only entered "kratom" into the substance verbatim field when a more specific product was known. We could not do a primary search by verbatim and it is possible that we missed some exposures that were not mapped to the kratom generic code. At the time of the query, the database was unlocked and subsequent queries may result in different counts. In the early phases, there were small numbers of cases, but looking at examples like the Diamond Shruumz™ outbreak, small numbers of novel products may spark investigations and further evaluation.

Conclusions

The substance verbatim field is a welcome addition to the database as it allows for the identification of substances at the national level that do not have an assigned product or generic code. The example described with 7-hydroxymytraginine demonstrates how the substance verbatim field can be used to identify exposures before the addition of a product code or additional exposures that were not coded after the addition of a product code. Future applications of this field involve the integration of natural language processing and trend analysis to improve early signal detection of outbreaks through identification of new substance verbatim values or spikes in specific substance verbatims.

Acknowledgements

America's Poison Centers® maintains the National Poison Data System® (NPDS), which houses deidentified records of self-reported information from callers to the country's Poison Centers (PCs). NPDS data do not reflect the entire universe of U.S. exposures and incidences related to any substance(s). Exposures do not necessarily represent a poisoning or overdose and America's Poison Centers is not able to completely verify the accuracy of every report. NPDS data do not necessarily reflect the opinions of America's Poison Centers.

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ORCID

James B. Leonard  <http://orcid.org/0000-0002-6444-8950>

Kaitlyn Brown  <http://orcid.org/0009-0009-0330-7829>

Jonathan M. Colvin  <http://orcid.org/0000-0002-3117-8333>

Data availability statement

Data are available from the National Poison Data System under a data use agreement.

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